

Thai Dang Khoa Tran

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PROFESSIONAL SUMMARY

Software engineer and MS Computer Science student (graduating December 2026) with 2+ years of full-stack experience building production systems in Java Spring Boot, Vue.js, and PostgreSQL for major Japanese retail chains. Builds data pipelines, serverless AWS apps, and RAG/LLM features, with tests at every level wired into CI.

EDUCATION

University of the Pacific, Stockton, CA

January 2025 - Present

MS Computer Science | GPA: 3.86 | Expected graduation: December 2026

University of Science, VNU - HCM, Vietnam

2018 - 2022

BS Information Systems

EXPERIENCE

Full-Stack Software Engineer

November 2022 - December 2024

UNICCS Co., Ltd — retail software provider; #1 core-system share among Japan's top 30 drugstore chains

- Architected full-stack features for TAURUS, a core retail ERP (purchase ordering, SKU-level inventory, departmental P&L) used by Don Quijote, Welcia, and Sundrug, with Java Spring Boot backends and JavaScript/Vue.js frontends.
- Reduced query response times by 40% through SQL tuning, indexing, and schema redesign on PostgreSQL and DB2.
- Cut a dashboard stored procedure's runtime from 10 minutes to 30 seconds by isolating the slow join and indexing the key field.
- Deployed and maintained services on AWS, automating build, deployment, and monitoring workflows with Shell scripts.

Tech: Java Spring Boot, JavaScript, Vue.js, PostgreSQL, DB2, SQL, AWS, Shell, HTML, CSS

Software Engineering Intern

December 2021 - June 2022

FPT Telecom — part of FPT Corporation, one of Vietnam's largest tech companies

- Built Node.js/Express REST APIs for real-time monitoring of POP stations across FPT Telecom's Southern Vietnam network.
- Developed React/Next.js dashboards with Highcharts, backed by MongoDB schemas designed for fast monitoring queries.

Tech: React.js, Next.js, Node.js, Express, MongoDB, Highcharts

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, C++, C#, SQL, HTML, CSS
- **Frameworks:** React.js, Next.js, Vue.js, Node.js, Express, Spring Boot, FastAPI, Tailwind CSS, Hono, D3
- **Databases:** PostgreSQL, MySQL, MongoDB, DB2, SQL Server, Oracle, ClickHouse, Redis
- **Testing & CI/CD:** Vitest, Playwright, unit testing, integration testing, end-to-end testing, GitHub Actions
- **Cloud & Tools:** AWS (Lambda, S3, CloudFront, SNS, IAM), Docker, Vercel, Git, GitHub, Shell, Claude Code
- **AI/ML:** PyTorch, Hugging Face Transformers, BERT fine-tuning, NLP, ONNX Runtime, RAG, LLM APIs (Groq)
- **Concepts:** OOP, Data Structures & Algorithms, RESTful APIs, Database Design, Data Pipelines/ETL, Serverless, Infrastructure as Code, Agile/Scrum, LLM Integration, Prompt Engineering, AI Agents

PROJECTS

US Weather Forecast Pipeline — Personal Project

[Live Demo](#) · [Source Code](#)

- Built an end-to-end data pipeline (NWS + Open-Meteo → MongoDB → ClickHouse → Redis → Hono API → React) processing 1.7M+ hourly observations and 3M+ forecast snapshots for 53 US cities since 2023, with incremental loads and data-quality checks.
- Deployed it on AWS as infrastructure as code (SAM): serverless collection and ETL on Lambda + EventBridge, S3 storage, CloudFront delivery, SNS alerts, and least-privilege IAM, all at \$0 on the free tier.
- Compared 4 weather models against a best-match baseline (ECMWF best: 2.0°F average error 1 day ahead) in a D3 dashboard with map drill-down and forecast replay.
- Used Claude Code to build and compare 3 versions of 5 key features, keeping the best (Redis caching cut p95 latency 12×); fixed SQL/NoSQL-injection and IAM security flaws; 209 automated tests in CI.

Tech: Node.js, Hono, TypeScript, React, D3, MongoDB, ClickHouse, Redis, Docker, AWS (Lambda, S3, CloudFront, SAM), Vitest

RAG-Assisted Natural Language to SQL System — Personal Project

[Live Demo](#) · [Source Code](#)

- Built and deployed a full-stack RAG assistant (React + FastAPI) that answers plain-English questions about any CSV: BERT routes intent, term-gated retrieval supplies business definitions, and gpt-oss-120b writes SQL, with gpt-oss-20b and rule-based fallbacks.
- Raised accuracy on definition-dependent questions from 0% to 95% on a held-out domain, with settings frozen before testing, within 2.5 points of an oracle; diagnosed why naive few-shot RAG lowered accuracy (99.2% → 93.3%): it matched topic, not SQL structure.
- Built a SELECT-only SQL safety layer that blocked 13 prompt-injection variants; all 120B benchmark runs cost under \$0.25.

Tech: Python, FastAPI, React, TypeScript, BERT, ONNX Runtime, Groq, SQLite, pytest, Vercel, Render

CineVibes — Personal Project

[Live Demo](#) · [Source Code](#)

- Built a full-stack movie platform with CineBot, an LLM assistant grounded in the app's MongoDB catalog via tool use; 57 tests in CI.

Tech: React, Node.js, Express, MongoDB, Vite, JWT Auth, Groq, Vitest, Playwright, GitHub Actions